

Factory Default

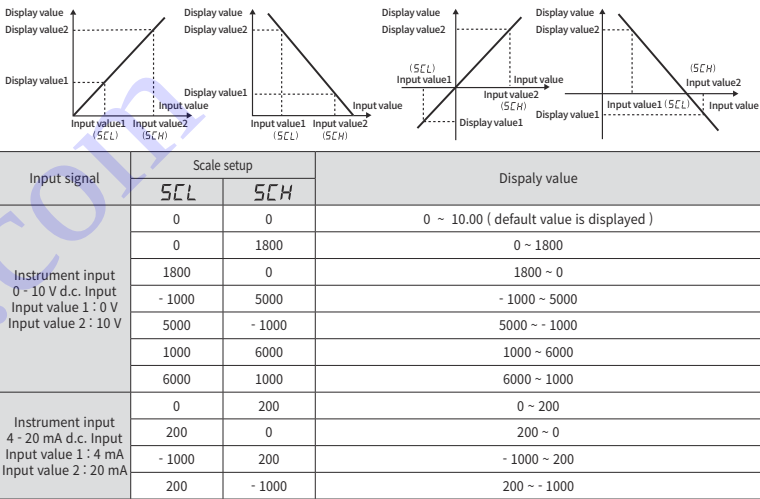
PARAMETER	MODE	MP3-DV	MP3-DA	MP3-AV (AVR)	MP3-AA (AAR)
PARAMETER 1	<i>1rS</i>	<i>500H</i>	<i>5R</i>	<i>500H</i>	<i>5R</i>
	<i>2HdP</i>	<i>1000</i>	<i>1000</i>	<i>1000</i>	<i>1000</i>
	<i>3LdP</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
	<i>4Rdt</i>	<i>0.2</i>	<i>0.2</i>	<i>0.5</i>	<i>0.5</i>
	<i>5Pdd</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
	<i>6SCH</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
	<i>7SCL</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
	<i>8dPP</i>	<i>0000</i>	<i>0000</i>	<i>0000</i>	<i>0000</i>
	<i>9PdH</i>	<i>oFF</i>	<i>oFF</i>	<i>oFF</i>	<i>oFF</i>
	<i>RLoC</i>	<i>oFF</i>	<i>oFF</i>	<i>oFF</i>	<i>oFF</i>
PARAMETER 2	<i>bRdr</i>	<i>00</i>	<i>00</i>	<i>00</i>	<i>00</i>
	<i>CbPS</i>	<i>96H</i>	<i>96H</i>	<i>96H</i>	<i>96H</i>
	<i>HHPL</i>	-	-	-	-
	<i>LLPL</i>	-	-	-	-
	<i>HSEt</i>	<i>5000</i>	<i>5000</i>	<i>5000</i>	<i>5000</i>
	<i>LSEt</i>	<i>2000</i>	<i>2000</i>	<i>2000</i>	<i>2000</i>
	<i>PSoC</i>	<i>oFF</i>	<i>oFF</i>	<i>oFF</i>	<i>oFF</i>
	<i>HYSt</i>	<i>01</i>	<i>01</i>	<i>01</i>	<i>01</i>

* *Rdr* and *bPS* only appears in the communication type

Parameter 1

Display	Meaning	Initial Setting	Setting Range	Note
<i>1rS</i>	DV Spec.	<i>500H</i>	0 <i>00SH</i>	Choose a measuring range of an input specification
			1 <i>SH</i>	
			2 <i>10H</i>	
			3 <i>50H</i>	
	DA Spec.	<i>5R</i>	4 <i>500H</i>	
			0 <i>SHR</i>	
			1 <i>20SHR</i>	
			2 <i>4-20</i>	
			3 <i>50SHR</i>	
			4 <i>0.5R</i>	
	AV Spec.	<i>500H</i>	5 <i>5R</i>	
			0 <i>SH</i>	
			1 <i>50H</i>	
	AA Spec.	<i>5R</i>	2 <i>500H</i>	
			0 <i>50SHR</i>	
			1 <i>0.5R</i>	
			2 <i>5R</i>	
<i>2HdP</i>	Magnification	<i>1000</i>	Max. Set Value : <i>1800</i> Min. Set Value : <i>0.500</i>	Sets the upper slope value of the input value.
<i>3LdP</i>	Error correction	<i>0</i>	Max. Set Value : <i>99</i> Min. Set Value : <i>-99</i>	Correct an error
<i>4Rdt</i>	Average input value measurement	<i>0.2</i>	Max. Set Value : <i>5.0</i> Min. Set Value : <i>0.1</i>	Set a time to measure an average input value
<i>5Pdd</i>	Peak detect delay time	<i>0</i>	Max. Set Value : <i>30</i> Min. Set Value : <i>0</i>	Set a delay time to detect a peak value
<i>6SCH</i>	High Scale	<i>0</i>	Max. Set Value : <i>9999</i> Min. Set Value : <i>-9999</i>	Set a higher limit of input (measuring) value
<i>7SCL</i>	Low Scale	<i>0</i>	Max. Set Value : <i>9999</i> Min. Set Value : <i>-9999</i>	Set a lower limit of input (measuring) value
<i>8dPP</i>	Dot Point	<i>0000</i>	<i>0000 0000 0000</i>	Set a position of a decimal point
<i>9PdH</i>	Auto holding	-	<i>oFF</i>	Set auto holding when detecting peak value
			<i>H-Hd</i>	
			<i>L-Hd</i>	
<i>RLoC</i>	Lock	-	<i>oN</i>	Use external hold function (E-Hd)
			<i>oFF</i>	
<i>bRdr</i>	Address	<i>00</i>	Max. Set Value : <i>99</i> Min. Set Value : <i>00</i>	Set Communication address and BPS ※ Display only RS485 communication output
<i>CbPS</i>	BPS	<i>96H</i>	<i>12H</i>	1200
			<i>24H</i>	2400
			<i>48H</i>	4800
			<i>96H</i>	9600
			<i>192H</i>	19200

Parameter 1 setting example)



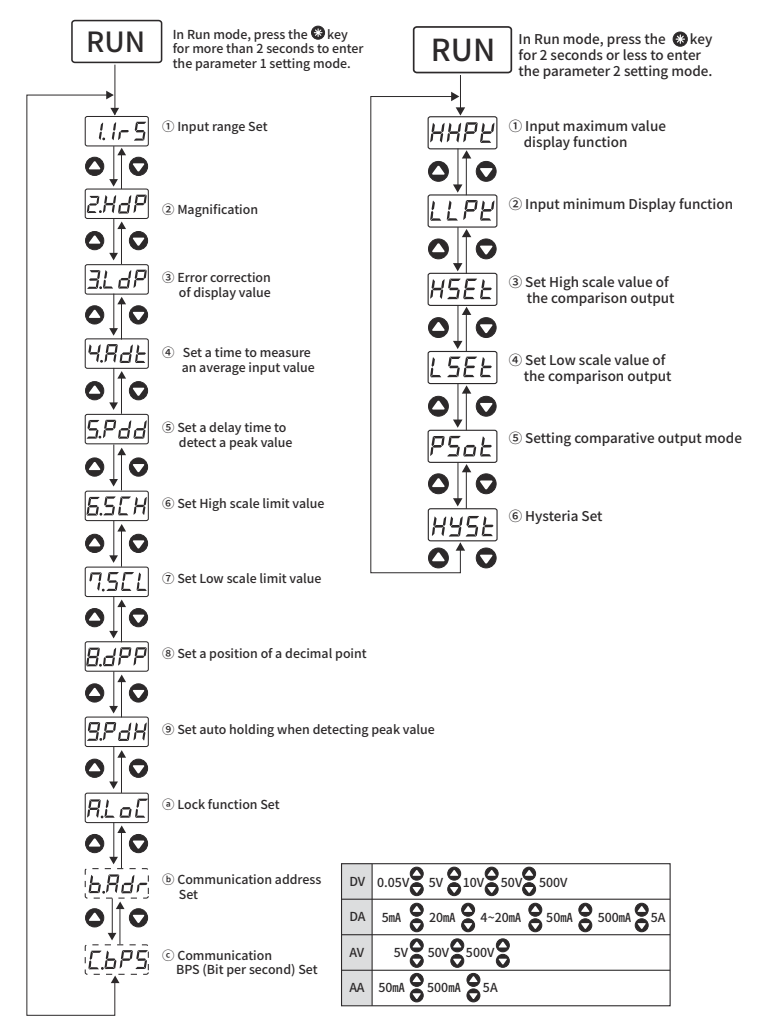
※ Built scale function, which can convert input signals into other values is available.
※ Rising, reversing and + - - display can be adjusted freely.

Parameter 2

Display	Meaning	Initial Setting	Setting Range	Note
<i>HHPL</i>	High Peak Display	-	Display value	Display max value among present input values.
<i>LLPL</i>	Low Peak Display	-	Display value	Display min value among present input values.
<i>HSEt</i>	Set High scale value	<i>5000</i>	Max. Set Value : + 9999 Min. Set Value : - 1999	Set High scale value of the comparison output
<i>LSEt</i>	Set Low scale value	<i>2000</i>	Max. Set Value : + 9999 Min. Set Value : - 1999	Set Low scale value of the comparison output
<i>PSoC</i>	Output Type Select	<i>oFF</i>	LL (LL.ot)	Set mode of comparison output
			HH (HH.ot)	
			LH (LH.ot)	
			HL (HL.ot)	
<i>HYSt</i>	Hysteresis	<i>01</i>	IL (IL.ot)	Set hysteresis of Comparison output
			Max. Set Value : 99 Min. Set Value : 00	

Parameter Setting Method

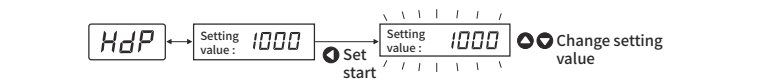
Parameter 1



DV	0.05V	5V	10V	50V	500V
DA	5mA	20mA	4-20mA	50mA	500mA
AV	5V	50V	500V		
AA	50mA	500mA	5A		

How to change Setting of Parameters

- In RUN MODE, press **F** for more than 2 seconds to enter Parameter 1 setting mode or for less than 2 seconds to enter Parameter 2.
- By using **0-9** keys, you can see one of the parameters in each parameter 1 or 2. If you are at one of the parameters, the parameter and set value is flickering repeatedly in the display.
- By using **F** you can begin with changing a setting value in the chosen parameter. Only the setting value in the chosen parameter is flickering after pressing **F** key. (When the default is 0, the number is flickering with only 0 th digit.)



Ex) When the set number is 0, the number is flickering with only 0th digit. To change 100th digit, press **F** key for 3 times. Each time you press The digit moves to the left, and the value of the selected digit flashes. When setup is complete **F** Press Key to enter parameter mode. At this time, the parameter and setting value are repeatedly blinked. To return to RUN mode press Key to return to RUN mode.

Parameter 1 Terminology Explanation

<i>1rS</i>	Choose a measuring range of an input specification
<i>2HdP</i>	Use to multiply the input measured value
<i>3LdP</i>	Correct an error by adding/subtracting to an input measured value. Ex) If you want to display 5000 but the measured value is 5010, then Ldp set -10 in parameter. The displayed value will be 5000. Default=0
<i>4Rdt</i>	It is difficult to measure an accurate value in a place of where an input measured value is varying too much. In this event, it is possible to display it as an averaged value by changing the cycle of the setting value. Ex) If set 2.0 in parameter then it will be displayed an averaged value for every 2 seconds after taking the values for 2 seconds. Default=0.2 Sec.
<i>5Pdd</i>	In an event of detecting max/min value, there is a possibility of detecting an erroneous value by an effect of the initial overvoltage and overcurrent. To prevent this matter, set delay time of detecting. Ex) If set 5 in parameter after connecting power source then it will detect max/min value after 5 seconds. When changing a setting value during an operating it will not affect detecting the present max/min value. Default=0
<i>6SCH</i>	Function which sets a higher limit of input (measured) value. (= display value A or B) SCALE function, Default=0
<i>7SCL</i>	Function which sets a lower limit of input (measured) value. (= display value B or A) SCALE function, Default=0
<i>8dPP</i>	Set a position of a decimal point. Default=000.0
<i>9PdH</i>	When detecting a peak value, you can use an automatic holding function. Default=OFF. Ex) if you choose H-Hd (Holding max value) in parameter then it will display max value among the input measured values, including the present value when the present value is less than the max value. In contrast, if the present value is higher than the max value then it will be displayed the present value by the automatic hold. The way of operating L-Hd hold (holding min value) is the same as the way of H-Hd hold (holding max value) does.
<i>RLoC</i>	This function is used to lock the parameter function of the panel meter. Ex) if parameter sets to be ON then it is impossible to set any parameters at this time.
<i>bRdr</i>	Set Communication Address. Default=00
<i>CbPS</i>	Set BPS Rate. Default=9600 bps

Parameter 2 Terminology Explanation

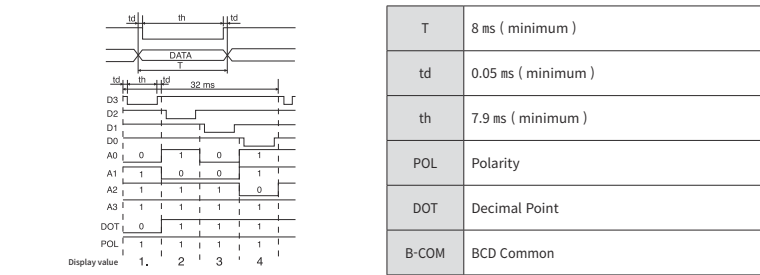
<i>HHPL</i>	Display max value among measured values of inputs.
<i>LLPL</i>	Display min value among measured values of inputs.
<i>HSEt</i>	Set a value of High Comparative Output. Default=5000
<i>LSEt</i>	Set a value of Low Comparative Output. Default=2000
<i>PSoC</i>	Select an operating mode of Comparative Output. Default=OFF
<i>HYSt</i>	Set hysteresis of Comparative Output. Default=01

BCD Static Output (Auxiliary Logic)

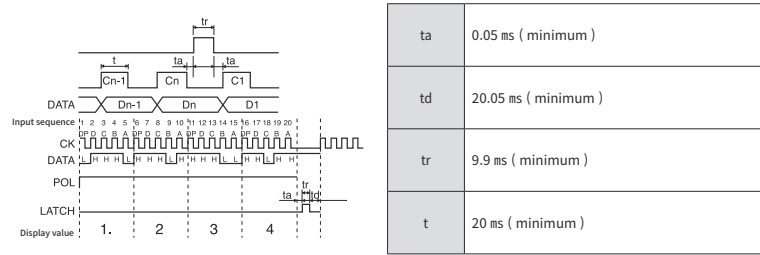
Wiring diagram

Pin NO.	Signal
1	DP1
2	DP2
3	POL
4	8 X 10 ⁴
5	8 X 10 ³
6	4 X 10 ³
7	4 X 10 ²
8	2 X 10 ³
9	2 X 10 ¹
10	1 X 10 ²
11	1 X 10 ¹
12	8 X 10 ²
13	8 X 10 ¹
14	4 X 10 ²
15	4 X 10 ¹
16	2 X 10 ²
17	2 X 10 ¹
18	1 X 10 ²
19	1 X 10 ¹
20	GND

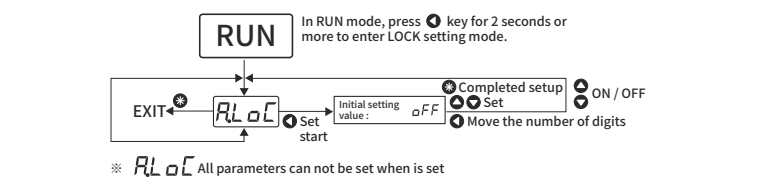
BCD Dynamic Output (Auxiliary Logic)



Low Speed Serial Output (Auxiliary Logic)



Parameter lock key



How to Set Slope

If you want to display 3600/min -1999 ~ 9999) then the model specification should be MP3-4-DV-X and an input should be 10V. The way of changing the parameter setting is written the below (Follow the steps 1 ~ 4).
1. By pressing **F** key for more than 2 seconds, enter Parameter 1 mode.
2. In *1rS* (Input Range) parameter, set *10H* range mode.
3. In *6SCH* (scale high limit) parameter, set *3600* by using shift, up and down keys.
4. In *7SCL* (scale low limit) parameter, set by using shift, up and down *0* keys.
Set the above and press Menu key again to go back to RUN MODE.

	Parameter 1		
Parameter	<i>1rS</i>	<i>SCH</i>	<i>SCL</i>
Set Value	<i>10H</i>	<i>3600</i>	<i>0</i>

Defaulting Set Values

- While pressing **F** key, press the key **F** → **F** → **F** → **F** then will be displayed. *EESt*
- At that time **F** press **F** key again then all the set values will be defaulted.
(※ If *RLoC* function is ON, it is impossible to be defaulted)

Error Display Code

<i>-HH-</i>	This will be displayed when it is higher than Max Range 9999 (for 4 digits model) or a negative number is appeared in Normal mode. Normal mode : SCH 0 or SCL 0
<i>OHFr</i>	This will be displayed when a measured input value is over max input range.
<i>HLEr</i>	A setting error will be displayed when a setting value of High Comparative Output is less than that of Low Comparative Output.

Retransmission Function (Auxiliary Output)

- RS485 Communication
- By assigning Address from 00 to 99 and by selecting modulation speed of Serial Transmission it is possible to transmit.
- Retransmission Speed (BPS) Selection Setting: (1200, 2400, 4800, 9600, 19200, 38400)
- Serial Communication
- Having a present displayed value generates signals of POL (polarity), DOT (decimal point), CK (CLOCK), DATA and LATCH in order to be easily done with connecting PLC and other processors.
- Current Output
- Generate 4 - 20mA d.c.for a present displayed value. (Resolution 12000)
- BCD Output (dynamic, Static)
- Generate a present displayed value as BCD type (D0, D1, D2, D3, POL (polarity) DOT (decimal point), A0, A1, A2, A3)
- PNP Output (Open Collector Output 12 - 24 V d.c. 50 mA)
- NPN Output (Open Collector Output 12 - 24 V d.c. 50 mA)
- RELAY Output (250 V a.c. 5 A below) 1a, 1b X 3

Auto Zero function

In the RUN mode, Press **F** → **F** buttons to display " *RLoC* " press. Auto - Zero the RUN mode, Press **F** → **F** buttons to display press, buttons to run Auto-Zero function instead of 0 value, the different value is displayed due to input source err, Auto Zero function adjusts 0 value automatically

Comparative Output Mode (*PSoC*)

